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Environmental Engineering (EE);
Colour and marking of DC cable and connecting devices

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Foreword

This final draft ETSI Standard (ES) has been produced by ETSI Technical Committee Environmental Engineering (EE), and is now submitted for the ETSI Membership Approval Procedure (MAP).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Introduction

The present document has been developed to assist the ICT industry in their development and wider use of DC powering solutions above the DC voltage range mainly used until now for Telecommunication network equipment (-48 VDC ETSI EN 300 132-2 [7]).

The present document has been jointly developed by ETSI TC EE and ITU-T Study Group 5 in order to ensure equivalent technical content. It is published respectively by ITU as Recommendation ITU-T L.1203 [1] and ETSI Standard ETSI ES 203 408.

DC power has been the main source of power for the Telecommunication industry since the very beginning as it is efficient, reliable and easily scalable.

This solution is a well proven arrangement that has been used for over a century in almost all telecommunication network equipment sites in cohabitation with AC for servers and other buildings services (lighting, facility plugs, lifts, etc.).

It is recognized that DC power has many potential applications further to those for ICT equipment and that such applications could benefit from the standards defined in ITU/ETSI. It is also recognized that further standardization activities on LVDC are now starting in other standardization organizations such as IEC. Thus ITU-T/ETSI will pursue cooperation with such organizations to reach maximum international alignment.

ETSI and ITU-T have standardized the widespread DC power interface at inputs of Telecom/ICT equipment used in Telecom networks, Data centers, and customer sites. The legacy interface for billions of equipment is the SELV voltage range so called -48 VDC, where the positive pole is grounded as defined for example for interface "A" in ETSI EN 300 132-2 [7]. Other complementary ETSI standards define a whole DC distribution including earthing and bonding such as ETSI EN 300 253 [9].

Higher voltages than -48 VDC are already widely used in Telecom networks for remote powering of signal repeaters on copper or optical Telecom transmission lines, with a usual operation in symmetrical voltage up to ±200 V in RFT-C or RFT-V telecom remote power feeding standardized in ETSI EN 302 099 [i.1].

The higher power density of ICT equipment and overall site power in MW for Data centers have led to the use of higher voltage levels, firstly in AC single or 3 phases up to 400 VAC at the input of servers. The need for resilience, energy efficiency and cost optimization has pushed the introduction of the up to 400 VDC power feeding interface standardized in ETSI EN 300 132-3 [8] and in Recommendation ITU-T L.1200 [2]. This is now available on many servers and Telecommunication network equipment.

As there is a progressive introduction of the new DC power interface in cohabitation with existing -48 VDC and AC distribution, ETSI EN 301 605 [10] has been introduced to ensure safety and proper operation, especially in mass bonding of EMC sensitive equipment and correct earthing connection to the building structure earthing network.

There is also a requirement for Telecom/Data center operators, equipment manufacturers, installers and end customers to clarify the colour and marking of cables in their installations. This avoids confusion and errors between the different AC and DC power interfaces and distribution equipment used in the building and inside ICT systems.

Considering the user side, and very long background experience in use of DC of the Telecom sector, the present document makes recommendations directly contributing to safety principles laid down in IEC Guide 104 [i.9] and ISO/IEC Guide 51 [i.10]. It accelerates and encourages the preparation of international standards of cable identification intended for use by manufacturers, users or certification bodies.

With reference to safety, it is important there is no ambiguity with respect to cable colours and identification. The present document therefore recommends harmonisation with existing IEC standards including IEC 60445 [3] and IEC 60050-195 [i.2], allowing a standardized industry approach.

For useful information, colours of cables inside equipment are considered in annex A as DC is already widely used for a multitude of applications, e.g. ICT equipment, car and trucks electric circuits, solar PV, etc.

1 Scope

The scope of the present document is to define common practice for identification of cabling or parts of the DC power distribution for the Telecom sites (buildings/rooms) feeding ICT and facilities equipment (power plant, cooling, building access systems, monitoring, etc.).

This applies to 400 VDC cabling for ICT equipment and environment equipment using interface defined in ETSI EN 300 132-3 [8] or Recommendation ITU-T L.1200 [2] and avoids confusion between the DC colour and marking of up to 400 VDC distribution (interface A3) and the other power interfaces identification: -48 VDC distribution interface A, AC interface A1 and uninterrupted AC given by inverters or UPS systems used in the building.

To ensure safety, it is important there is no ambiguity with respect to cable colours and identification, The present document therefore recommends harmonisation with existing IEC standards including IEC 60445 [3] and IEC 60050-195 [i.2], allowing a standardized industry approach.

The Recommendation defines:

- requirements for the colour and identification of separate wires used in DC distribution systems and installations, and as much as possible inside equipment between A or P interface and end use;
- marking and identification of DC multiwire cables, connectors and any associated equipment for DC power distribution;
- marking identification for sui-distribution boards and interconnection boxes;
- harmonisation with IEC 60445 [3].

For DC wires in multiwire cables, recommendations are given for harmonization with separate individual wire requirements.

Recommendations are given on other distribution items (wires, cables and interconnection items) including the functional earthing and bonding arrangement for distribution outside and inside equipment to avoid any confusion with DC distribution identification of the present document.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

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The following referenced documents are necessary for the application of the present document.

[1] Recommendation ITU-T L.1203: "Colour and marking identification of up to 400VDC power distribution for ICT systems".

[2] Recommendation ITU-T L.1200 (2012): "Direct current power feeding interface up to 400 V at the input to telecommunication and ICT equipment".

[3] IEC 60445:2021: "Basic and safety principles for man-machine interface, marking and identification - Identification of equipment terminals, conductor terminations and conductors".

[4] IEC 60757:2021: "Code for designation of colours".

[5] IEC 60417:2026 DB: "Graphical symbols for use on equipment".

[6] IEC 60617:2026 DB: "Graphical symbols for diagrams".

NOTE: IEC DB documents refer to Database of Graphical Symbol. Some are joint with ISO e.g. IEC 60417 and ISO 7000.

[7] ETSI EN 300 132-2 (V2.8.1): "Environmental Engineering (EE); Power supply interface at the input of Information and Communication Technology (ICT) equipment; Part 2: -48 V Direct Current (DC)".

[8] ETSI EN 300 132-3 (V2.3.1): "Environmental Engineering (EE); Power supply interface at the input of Information and Communication Technology (ICT) equipment; Part 3: Up to 400 V Direct Current (DC)".

[9] ETSI EN 300 253 (V2.2.1): "Environmental Engineering (EE); Earthing and bonding of ICT equipment powered by -48 VDC in telecom and data centres".

[10] ETSI EN 301 605 (V1.1.1): "Environmental Engineering (EE); Earthing and bonding of 400 VDC data and telecom (ICT) equipment".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI EN 302 099 (V2.3.1): "Environmental Engineering (EE); Powering of equipment in access network".

[i.2] IEC 60050-195 (2021): "IEC Terminology International Electrotechnical Vocabulary - Part 195: Earthing and protection against electric shock".

[i.3] IEC 60050-826 (2022): "International Electrotechnical Vocabulary - Part 826: Electrical installations".

[i.4] BS 7671/A4 (2026): "Requirements for Electrical Installations. IET Wiring Regulations".

[i.5] HD 308 S2:2001: "Identification of cores in cables and flexible cords".

[i.6] HD 324 S1 (withdrawn) (1977): "Colored insulation and bare wire identification package".

[i.7] HD 384.5.514 (withdrawn): "Identification, including 514.3: Identification of conductors".

[i.8] IEC 60364-1 (2025): "Low-voltage electrical installations - Part 1: Fundamental principles, assessment of general characteristics, definitions".

[i.9] IEC Guide 104: "The preparation of safety publications and the use of basic safety publications and group safety publications".

[i.10] ISO/IEC Guide 51: "Safety aspects - Guidelines for their inclusion in standards".

[i.11] ISO 19642 (2023): "Road vehicles. Automotive cables".

[i.12] ISO 2574 (1994): "Flying machine electric cable - Identification marking".

[i.13] ISO 9247 (1990): "Ground work machine - electric wire and cables -- identification and marking principles".

[i.14] IEC EN 60204-1 (2018): "Safety of machinery -Electrical equipment of machines -Part 1:General requirements".

[i.15] IEC 60304 (1982): "Standard colours for insulation for low-frequency cables and wires".

[i.16] Void.

[i.17] Void.

[i.18] Void.

[i.19] IEC 60446 (withdrawn) (2007): "Basic and safety principles for man-machine interface, marking and identification - identification of conductors by colours or alphanumerics".

[i.20] CCSA YD/T 2378 (2020): "240V Direct Current Power supply system for Telecommunications".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the following terms and definitions apply:

cable identification: colour, graphical symbol, text in alphanumeric notation located on or adjacent to wire or cable terminals or junction boxes

electrical equipment: item used for purposes like generation, conversion, distribution or utilization of electric energy (e.g. electrical machines, transformers, switch gear and control gear, measuring instruments, wiring systems, current-using equipment, etc.) (IEC 60050-826 [i.3])

IT (Isolation Terra): Isolation Terra earthing as defined in IEC 60364-1 [i.8].

NOTE: The two letters IT are coding one of the 3 families of electrical distribution where connection to earth is made through high resistance or impedance device i.e. HRMG in ETSI EN 301 605 [10].

line conductor: AC phase conductor or DC pole conductor (deprecated) conductor which is energized in normal operation and capable of contributing to the transmission or distribution of electric energy but which is not a neutral conductor (IEC 60050-195 [i.2])

TN-S DC: Distribution earthing arrangements as defined in IEC 60364-1 [i.8] and ETSI EN 301 605 [10] where earth and negative (DC) wires are separated

3.2 Symbols

For the purposes of the present document, the following symbols apply:

48 VDC	-48 Volt Direct Current voltage as defined in ETSI EN 300 132-2 [7]
400 VDC	up to 400 Volt Direct Current voltage as defined in Recommendation ITU-T L.1200 [2]
A (interface)	ICT/Telecom equipment -48 VDC power interface as defined in in ETSI EN 300 132-2 [7]
A3 (interface)	ICT/Telecom equipment up to 400 VDC power interface as defined in ETSI EN 300 132-3 [8]
P	ICT/Telecom equipment up to 400 VDC power interface as defined in Recommendation ITU-T L.1200 [2]

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AC	Alternative Current
DC	Direct Current
EMC	Electro Magnetic Compatibility
HRMG	High Resistance Mid-point Grounding

ICT	Information and Communication Technologies
LVDC	Low Voltage Direct Current
MW	MegaWatt (unit)
OFC	Oxygen Free Cable
PE	Protected Earthing conductor
PEL	Protective Earthing conductor and Line conductor

NOTE: As defined in IEC 60050-195 [i.2].

PEN	Protective Earthing conductor and Neutral conductor

NOTE: As defined in IEC 60050-195 [i.2].

RA	Remote Access
RFT-C	Remote Feeding Telecom-Current
RFT-V	Remote Feeding Telecom-Voltage
SELV	Safe Extra Low Voltage
UPS	Uninterruptable Power Supply
VAC	Voltage Alternating Current
VDC	Voltage Direct Current

4 Electrical Distribution identification coverage

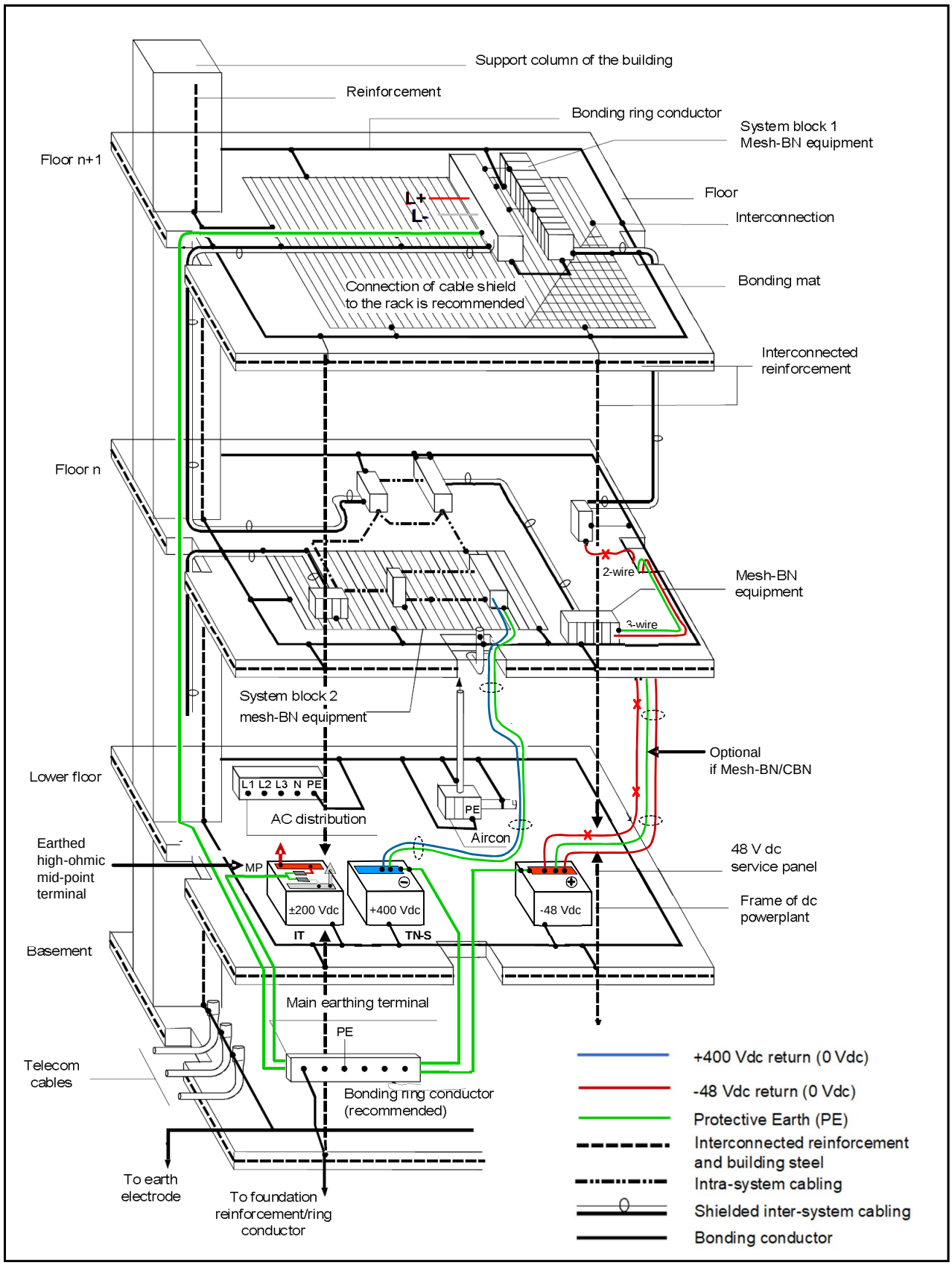
The DC Electrical Distribution needs marking at the following different points:

- DC distribution lines (wires and cables)
- DC cabling inside DC power sources
- DC outputs from DC power systems with protective devices
- DC sub-distribution boards (with protective devices)
- DC interconnection boxes
- DC power plugs strips inside ICT equipment rooms and cabinets

Typical building and room distribution are provided in the following drawings:

- Building distribution with interface A (-48 VDC), AC grid and inverter outputs power and interface A3 or P (up to 400 VDC). The interface shall comply with:
  - for interface A ETSI EN 300 132-2 [7];
  - for interface A3 ETSI EN 300 132-3 [8];
  - for interface P Recommendation ITU-T L.1200 [2].

  Figure 1 shows such building where distribution DC and Protective Earthing-[PE] bonding shall comply with ETSI EN 301 605 [10] and ETSI EN 300 253 [9].
- Example of power feeding distribution of DC from centralized or decentralized permanent power supply at building or technical room level (figure 2).



NOTE: The white cable is representing by light grey colour for ±200 VDC.

Figure 1: DC distribution at building level from DC source to ICT/telecom systems

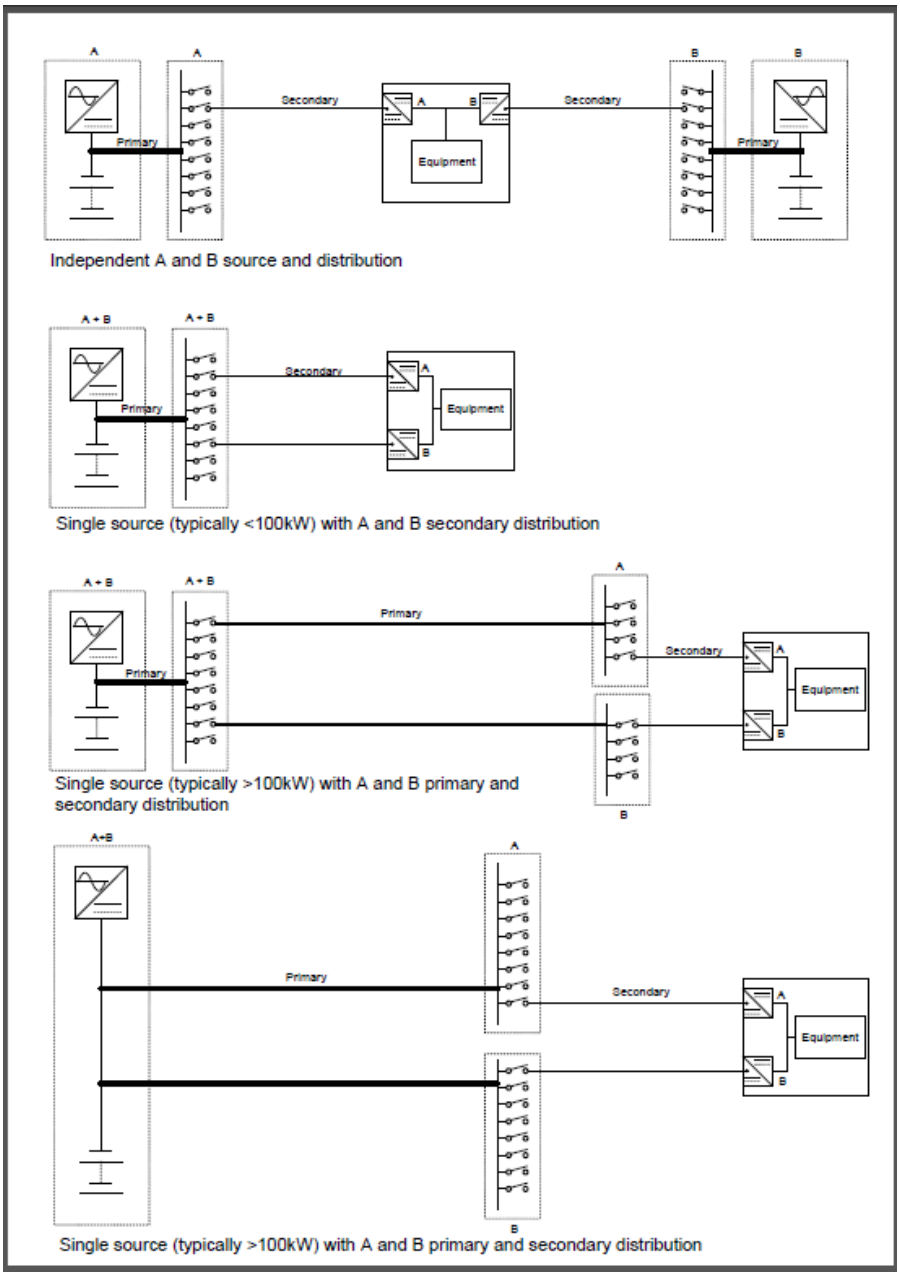


Figure 2: Example of different DC architectures and distributions at building or technical room level from DC source to ICT/telecom equipment

5 Identification principle

5.1 General rules

The identifying colour, graphical symbols or alphanumeric notations used for the identification of conductors and terminals shall follow the requirements as defined in IEC 60445 [3].

Where letters and/or numerals are used, letters shall be upper case Latin characters, numerals shall be Arabic numerals.

The identifying colour, graphical symbol or alphanumeric notation shall be located on, or adjacent to, the corresponding terminal.

The identification of colour shall be used at the terminations and shall extend throughout the full length of the conductor from the DC source to interface A3 or P, either by the integral colour of the insulation or by using colour markers and identification labels applied at regular intervals. This includes within the equipment cabinet to the ICT or facilities equipment (Cooling, Power Conversion, Sub Distribution, etc.).

There shall be no confusion between the different identification modes i.e. wire insulation colour or coloured sleeve, interconnection device colour and graphical symbol or text. For identification of conductors, only colours and code among those permitted in IEC 60445 [3] and IEC 60757 [4] shall be used.

DC cabling inside power source equipment is recommended to use the same colour marking and identification labels as defined in the power feeding distribution.

It is accepted to have different identification marking solutions within an installation (insulation colour, markers, sleeves, symbol and text, etc.), provided they are compliant to the present document. In that case, wires, conductors and cable terminations shall use these clear marking solutions.

The colour and marking identification requirements shall be applied to new installations at building level, room level and equipment level. There is no specific requirement to update the colour marking and identification labels on an existing installation which is not changed or modified as part of installation work; however it is important to ensure the identification of all wires, conductors and terminations is clear. When making an extension or alteration to an existing installation, all wires, conductors and terminations shall be marked so there is no ambiguity. Where more than one identification method is used, the correlation between the two different identification methods shall be fully documented.

5.2 Colour and marking of up to 400 VDC conductor wires and cables for A3 or P interface

Each conductor shall be identified with the colour marking at the termination and throughout its full length. For colour marking, where possible the preference is to use the integral colour of the cable insulation. Alternatively, colour markings and identification labels shall be applied at regular intervals throughout the full length of the conductor. At terminations, colour marker and identification labels shall be applied to the cable as close to the point of termination as possible. This is to avoid confusion where there is restricted access due to excess cables being present.

NOTE: Text marking should be used as much as possible as it is non ambiguous and is clear.

For bare conductors the colour marking and identification labels shall be applied at termination and connection points.

If the separate wire insulation is identified using the colour defined in table 1, no other colour markers are required on the wire.

Single wire:

- Single wires of a new installation shall be identified using the colour markings and identification labels as defined in table 1.
- In the case of an extension or alteration of an existing installation, identification labels respecting table 1 shall be applied to all conductors that are not yet using colours defined in table 1. The use of identification markers such as sleeves or rings is possible.

Multiwire cables:

- If the wire inside a multiwire cable does not have the correct colour as defined in table 1, colour markings and identification labels shall be added respecting table 1 with describing text, e.g. by sleeves or rings at both ends.
- For single and multiwire cable, the functional identification of the wire by labels or rings with alphanumeric character shall be the same.

The colour and marking requirements do not apply for existing equipment distribution and cabling, unless extensions and alterations are made as part of the installation work.

It shall be applied on completely new installation and equipment.

Additional markings to table 1, for example alphanumerical and colour marking, are permitted provided they do not introduce any ambiguity for users.

NOTE 2: Table 1 does not define terminals/conductor markings used for earthing with their different purposes of earthing (divided as two basic concepts of protective earthing and functional earthing), but requires to use colours and marking of existing IEC standard. In compliance with IEC 60445 [3], the requirements for functional earthing should be defined by the manufacturer or the relevant product committee and should be specified within the documentation of the equipment. Other requirements for handling EMC issues are defined in ETSI EN 301 605 [10].

Other conductor types:

Identification by colour or marking is not required for:

- concentric conductors of cables;
- metal sheath or armour of cables when used as a protective conductor;
- bare conductors where permanent identification is not practicable;
- extraneous-conductive-parts used as a protective conductor;
- exposed-conductive-parts used as a protective conductor.

Table 1: Requirements for up to 400 VDC colour and marking for separate wires and for wire in multiwire cables (for 48 VDC and 60 VDC refer to table 2)

DC lines	Colour marking	Conductor Identification	Terminal Identification	Standard bases
Circuit or installation downstream and upstream to interface A3 or P, external or internal to the equipment	Integral to wire insulation material or on markers (sleeves, rings, etc.)	Text on cable label or attached to connectors	Alphanumeric symbol at the termination	For colour and graphical symbols
Positive line HRMG	Red (see note)	L+ and voltage level +190 VDC	+	IEC 60617 [6] IEC 60417 [5] IEC 60757 [4] IEC 60445 [3]
Positive line	Red (see note)	L+ and voltage level +380 VDC		
Negative line HRMG	White (see note)	L- and voltage level -190 VDC	-	IEC 60617 [6] IEC 60417 [5] IEC 60757 [4] IEC 60445 [3]
Negative line connected to PE	Blue (Light blue) (see note)	L- and voltage level 0 VDC		
NOTE: For multiwire cables and alteration of existing installation with separate wires (e.g. for extension), when the insulation colour of a wire is different from the one of table 1, identifiers should be used (e.g. sleeves or rings).				

It is suggested to use text marking with +190 VDC, -190 VDC or +380 VDC, 0 VDC which can help to clearly distinguish the two different earthing systems.

5.3 Additional recommendations for other DC voltage distribution cabling excluding up to 400 VDC interface A3 or P

Table 2 adds some recommendations for -48 VDC and -60 VDC distribution colour and marking, to avoid confusion with up to 400 VDC distribution. The -48 VDC corresponds to interface A of ETSI EN 300 132-2 [7].

It should be applied to other LVDC voltage conductors, excluding up to 400 VDC as defined for interface A3 or P.

Table 2: Recommendation for -48 VDC and -60 VDC colour and marking

DC lines	Colour marking	Conductor Identification	Terminal Identification	Standard bases
Circuit or installation downstream and upstream interface A, external or internal to the equipment	Integral to wire insulation material or on markers (sleeves, rings, etc.)	Text on cable label or attached to connectors	Alphanumeric symbol at the termination	For colour and graphical symbols
Positive Line Connected to PE	not defined (see note)	M (0 VDC)	M	IEC 60617 [6] IEC 60417 [5] IEC 60757 [4] IEC 60445 [3]
Negative Line	not defined (see note)	L- (-48 VDC or -60 VDC)	-	IEC 60617 [6] IEC 60417 [5] IEC 60757 [4] IEC 60445 [3]
NOTE: Colour markings are not defined as many national and industry standards are already widely deployed.				

6 DC distribution devices additional colour and marking

6.0 Colour and marking of up to 400 VDC distribution devices, other than wires and cables

Table 3 gives requirements for colours and marking of electrical boxes.

Table 3: Requirement for colours and marking electrical boxes

Interface	Material or label background colour (see note 1)	External Text writing colour on background	Single voltage Compartment Text	Standard bases
	Box external marking	Box external marking		For colour and graphical symbols
A3 (400 VDC)	different from A and A1	(see note 2)	DC voltage level: e.g. 380 VDC Earthing arrangement: HRMG (IT) or L- to earth (TN-S)	IEC 60617 [6] IEC 60417 [5] IEC 60757 [4] IEC 60445 [3]
A (-48 VDC)	different from A3 and A1	(see note 2)	DC voltage level: -48 VDC Earthing arrangement: L+ to earth	IEC 60617 [6] IEC 60417 [5] IEC 60757 [4] IEC 60445 [3]
A1 (AC)	different from A3 and A	(note 2)	AC (nominal voltage) Voltage level: e.g. 230 VAC Earthing arrangement: IT, TN-S, etc.	IEC 60617 [6] IEC 60417 [5] IEC 60757 [4] IEC 60445 [3]
NOTE 1: For the background colours, it can be integral colours of the box plastic material or painted colour, or an added wide label using the defined back and text colours of table 3 on the new installation. NOTE 2: Symbols for DC and AC of IEC 60417 [5]. The chosen colour should be very readable on the chosen background colour.				

6.1 General requirements

In general, within a building the distribution cables may not run directly from the source to the equipment (see figure 2). This may be due to the building layout, room layout, containment routes, maintenance and extension works, and reliability optimization, etc.

Connection boxes or distribution boards are used in Data centers or in Telecom network sites in many places along the electrical distribution e.g. within floor voids, in wall mounted configurations, at heads of Telecom equipment or in suite lines, etc.

There should not be mixed distribution of cables of different voltage levels e.g. -48 VDC, AC and 400 VDC inside the same box or there should be separated compartments for each with clear marking and identification.

NOTE: National electrical safety rules have to be applied for this matter.

In addition to conductor colour and marking, there shall be additional identification to avoid any confusion [6] between -48 VDC, 400 VDC, and AC outputs at level of cabinets, racks or shelf and sui-blocks or modules inside.

Interconnection arrangements with different voltage ranges and types shall have different colours for easy identification. The table 3 requirement shall be applied for up to 400 VDC distribution (A3) and should be applied for -48 VDC (A) and AC (A1) distributions.

6.2 DC outputs from DC power systems with protective devices

The colours and marking should be the same as for DC interconnection boxes in table 3:

- Writing and background colours on labels
- The frames itself should be of the background colour of the labels (e.g. white, orange, blue)

- Marking of terminals and voltage

There shall not be mixed distribution of -48 VDC, 400 VDC, AC in the same mechanical module without proper insulation separators.

6.3 DC sub-distribution boards (with protective devices)

The colour and marking should be the same as for DC power systems outputs.

There shall not be mixed distribution of -48 VDC, 400 VDC, AC in the same mechanical module without proper insulation separators.

6.4 DC power plugs strips inside ICT equipment rooms and cabinets

The colour and marking should be the same as for DC power systems outputs.

A coloured round circle around each socket using respective colour of table 3 for respective voltage type could be used rather than full plastic colour.

7 Durability of Marking and reading errors limitation

All clause 7.1 of IEC 60445 [3] shall be applied.

Annex A (informative): Review of Standards or common use of colour and marking of distribution cabling in AC and DC in buildings and equipment

A.1 Recommended colours for wires in IEC 60757 and IEC 60445

The recommended colours of wires are given in Annex A of IEC 60757 [4], table A.1 - Recommended colours and code.

The following are preferred:

- Black BK, Brown BN, Red RD, Orange OG, Green GN, Yellow YE, Blue BU, Violet VT, Grey GY, White WH, Pink PK, Gold GD, Turquoise TQ, Silver SR.

IEC 60445 [3] permits the following colours for identifying conductors:

- Black, Brown, Red, Orange, Green, Yellow, Blue, Violet, Grey, White, Pink, Turquoise.

The colours green and yellow on their own are only permitted where confusion with the colouring of the green/yellow protective conductor is unlikely. Combinations of the above colours are permitted, but green and yellow should not be used in any of these combinations other than as green/yellow for the protective conductor.

If a circuit includes a neutral or midpoint conductor, then it should be identified by a blue colour (preferably light blue). Light blue is the colour used to identify intrinsically safe conductors and should not be used for any other type of conductor.

A.2 Other Norms or specifications

A.2.0 Introduction

Other activity sectors have standardized cabling and marking for their specific sector:

- The International Standard IS0 19642 [i.11] specifies the dimensions, test methods, and requirements for single-core 60 VDC cables intended for use in road vehicle applications where the nominal system voltage is lower than 60 VDC or 25 VAC. It also specifies additional test methods and/or requirements for 600 VDC cables intended for use in road vehicle applications where the nominal system voltage is in the range between 60 VDC or 25 VAC to 600 VDC or VAC. It also applies to individual cores in multi-core cables. It does not define colours of wire.

A.2.1 Single and bicolour cables used in transportation sector

To avoid any confusion, it is important to know that some industry sectors such as transportation (car, vans, boats, planes) are defining specification of single or bicolour wires for specific use of DC power e.g. motor sensors or auxiliary circuits. Figure A.1 is giving examples of bicolour wire insulation.



Figure A.1: Example of bicolour wire insulation

A.2.2 Cables used in flying aircrafts, ground works machines and boats

ISO 2574 [i.12] provides Flying machine electric cable identification marking and ISO 9247 [i.13] provides Ground work machine electric wire and cables identification and marking principles.

NOTE: The ship and boat sector also has specifications.

A.2.3 Colour of personal computer power cables

There is a common practice for colours cabling mostly used in "ATX" design of personal computer tower as defined in table A.1.

Table A.1

Wire colour	Function	Wire colour	Function
Black	ground power	Green	Power on
Orange	+3,3 V power	Purple	+5 V standby
Red	+5 V power	Grey	Power good
Yellow	+12 V power	Brown	+3,3 V sense
Blue	-12 V power	White	-5 V

A.2.4 Machine Cabling

IEC EN 60204-1 [i.14] standard on electric machines cabling was imposing light blue for neutral if the equipment has a DC circuit and dark blue for the negative pole in order to better discriminate AC and DC circuits.

A.2.5 Family of cables coding

This clause is giving indication on cable type codification and showing that transparent insulation is used in some case. Each type of wire or cable has a coding for a wide variety of use as seen in following examples:

Fixed Installations and electric board cabling

H07VU or VR, H07VK cable

- H: harmonised
  - 07: max voltage 700 volt.
    - V: Polyvinyl Chloride (PVC).
    - U: single core copper.
    - R: hard multicore conductor.
    - K: flexible multiwire.

U1000R2V cable

- 2V: double insolation PVC.

Codification (G or X) indicates ground cable, number of wires and section.

- Example: U1000R2V 3G 1,5 mm².
- U1000R2V 3 × 1,5 mm² means 3 wires, without ground yellow/green.

H03VVF/H05VVF

- VV: insolation with PVC material.
  - 03: < 1 mm² (2 wires).
  - 05: > 1 mm² (2 to 5 wires, for single or 3 phases).
  - F: soft wire.

H03VVH2F

- H2F: harmonised with 2 soft wires.

H07RNF

- Elastomeric insulation - (Neoprene).

H03RTF

- **Transparent** or sleeved copper for lighting, or other household device.

Hi-fi Wires

- For audio equipment including in vehicles and home cinema.
- Blue, red, black, white and **transparent**.
- Section: 0,75 to 6 mm².

Better in OFC (Oxygen Free Cable), less oxidized.

A.3 Building distribution cable colour and marking standards

A.3.1 Introduction

The standard about colour cable has changed with time from IEC 60446 [i.19] withdrawn to new IEC 60445 [3].

NOTE: The IEC 60446 standard Third edition 1999-02 of IEC 60446 [i.19] has been withdrawn but was giving information on the "Basic and safety principles for man-machine interface, marking and identification - Identification of conductors by colours or numerals".

The scope was focused on general rules for the use of certain colours or numerals to identify conductors with the aim of avoiding ambiguity and ensuring safe operation. These conductors may be applied in cables or cores, busbars, electrical equipment and installations. It is based on the principles given in IEC Guide 104 [i.9] and ISO/IEC Guide 51 [i.10].

A.3.2 Low frequency cable and wire reference insulation colour in IEC 60304

There has been harmonization in AC building distribution since 1970 towards conductor insulation colours and jackets of optical cables. The allowed colours suggested for standardization are proposed in IEC 60304 [i.15] defining the thermoplastic insulation for low-frequency cables and wires and the colours to be used as follows:

- **Light blue:** neutral.
- **Red or brown:** phase.
- **Bicolour Yellow and green:** earth.

In addition, some use of colours are preferred but not required:

- **Orange:** phase at the output of a receiver, e.g. phase in red or brown is connected to a lighting bulb and the other wire is connected to the interrupter with an orange wire (and not red).
- For "shuttle" line between 2 inverters of a forth and back circuit: all colours excepted the previous one and earth colour are allowed (white, grey).

A.3.3 IEC 60445

A.3.3.0 General information on IEC 60445

The standard IEC 60445 [3] is providing basic and safety principles for man-machine interface, marking and identification of equipment terminals, conductor terminations and conductors.

The old colours used in Europe are reminded.

A.3.3.1 Old Cable Colour Code reminder

Power cable insulation is normally colour coded so that phase, neutral and earth conductors can be easily identified. These colour codes vary with region and/or country as can be found in IEC 60445 [3].

A.3.3.2 Harmonized Cable colours

Identification colours of cores in cables have been subject to developments that results in the harmonization document HD 308 S2 [i.5].

These rules do not apply to conductors used in the materials and sets assembled at the factory although compliance is strongly recommended. A short extract of HD 324 S1 [i.6] is showing the historic colour choices for AC wires in cables.

A.3.3.3 Marking and alteration for cohabitation of harmonized and old Cable colours

This clause is useful to understand how to manage the cohabitation of old and new colour and marking in installation.

For information, old national habits are reminded. These cables are still widely present in existing installations.

In 2004, the requirements of BS 7671 [i.4] were harmonized with HD 384.5.514 [i.7] (now withdrawn) and an Amendment No 2: 2004 (AMD 14905) to BS 7671 [i.4] (now withdrawn) was implemented for the harmonized cable core colours and the alphanumeric marking of the following standards based on HD 308 S2 [i.5], IEC 60445 [3] and IEC 60446 [i.19].

Guidance was provided on marking at the interface between old and harmonized colours and marking general guidance on the colours to be used for conductors.

BS 7671 [i.4] was also modified to allow suitable methods of marking connections by colour (tapes, sleeves or discs), or by alphanumeric' s (letters and/or numbers). Methods may be mixed within an installation. An addition or an alteration made to a single-phase installation need not be marked at the interface. The old cables correctly identified by the colour red for line and black for neutral, can be interconnected to the new cables correctly identified by the colour brown for line and blue for neutral. Where an addition or alteration is made to an installation wired in the old core colours with cable to the new core colours, unambiguous identification is required at the interface. Cores should be marked as follows:

- Neutral conductors: Old and new conductors: N.
- Line conductors: - Old and new conductors: L1, L2, L3.

A.3.4 DC coding evolution

A.3.4.1 Colours and marking of DC cables in the UK

In 2022, BS 7671 [i.4] Amendment 2 was released which resulted in a change to the colours used for the for the identification of -48 VDC cable marking (two-wire earthed DC power circuit) and 400 VDC cable marking (two-wire unearthed DC power circuit). BS 7671 [i.4], table 51 details the conductor marking at the interface for -48 VDC and 400 VDC installations:

1) -48 VDC system marking (two-wire earthed DC power circuit)

   Positive (of positive earthed) circuit: M Blue

   Negative (of positive earthed) circuit: L- White

2) Up to 400 VDC system marking (two-wired unearthed DC power circuit):

   Positive of two wire circuit: L+ Red

   Negative of two wire circuit: L- White

NOTE: BS 7671 [i.4] Amendment 4 was released in April 2026, there are no changes to table 51.

A.3.4.2 Colours and marking of DC cables in China

China has also defined colour coding for DC cable:

1) -48 VDC system marking:

   For China Telecom and Unicom:

   L+ Red

   L- Blue

   For China mobile:

   L+ Red

   L- Light Blue

2) Up to 400 VDC system marking:

   For China Telecom and China mobile and in CCSA standard YD/T 2378 [i.20]:

   L+ Brown

   L- Blue

   AC wire colours in China:

   Blue (N line)

Red, Yellow, Green (Live lines)

3) Ground wire for all:

For all operators in China:

Yellow and green

A.3.4.3 Colours and marking of DC cables in Republic of Korea

Different DC voltage cables do not share the same power distribution panel. Power distribution panels are identified by marking, the voltage level, on the external cover.

DC + (positive) cable is identified with red colour and DC - (negative) with blue colour.

1) -48 VDC:

- DC + (positive) Red
- DC - (negative) Blue

2) up to 400 VDC:

- DC + (positive) Red
- DC - (negative) Blue

Annex B (informative): Bibliography

ISO/IEC 81346-1 (2022): "Industrial systems, installations and equipment and industrial products structuring principles and reference designations -- Part 1: Basic rules".

History

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V1.1.1	December 2016	Publication	
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